

Cultural heritage knowledge graph and recommender system

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Abstract— This study utilizes Knowledge Graph Attention Network (KGAT) to embed cultural heritage ontology data, thereby creating a recommender system based on the similarity of the heritages. We build a cultural heritage graph using the ontology data and embed all nodes and edges in the graph as vectors. We then make a recommendation result based on the cosine similarity. Also, we propose a method to embed a new cultural heritage using existing embedding vectors without retraining the model, and hence effectively addressing the cold start problem. Experiments demonstrate that our system creates a list of relevant cultural heritages as recommendation and handle new items efficiently without additional training. This method offers a new perspective on cultural heritage, with potential future research integrating user information for more precise recommendations.

Keywords—Knowledge graph, Graph embedding, Knowledge graph attention network, cold start problem, cosine similarity

I. INTRODUCTION

Analyzing the relationships among various cultural heritages is complex. Knowledge graphs can structurally represent and analyze this information, and by embedding them we can make a recommender system based on the similarity of the heritages [2]. Knowledge Graph Attention network(KGAT) is a model designed to effectively embed the structural information of a knowledge graph. It learns the information of neighboring nodes to generate embedding vectors and uses an attention mechanism to assign higher weights to important neighbors, improving the performance.

This study aims to use the KGAT (Knowledge Graph Attention Network) model [3]— to embed cultural heritages and attribute information, creating a system that recommends similar cultural heritages.

In recommendation systems, cold start problem refers to the problem of insufficient information to recommend appropriate products when there are new communities, items, or users [1]. We propose a method to generate embedding vectors for new cultural heritages based on the pre-trained embeddings without retraining the model. This can save computational time and cost for retraining and further address the cold start problem.

II. METHODOLOGY

2.1 Datasets

In this study, we use an ontology dataset that contains information about cultural heritage. The ontology includes details such as the specifics of each item, housing institution, collection number, name, patterns and appearance (Table I). It provides 34 attributes for 11,914 cultural heritages. The cultural heritage knowledge graph uses entities from the ontology data, with relations like ‘~. has pattern’ and ‘~ is housed at’.

2.2 Knowledge Graph

In this study, we structured the KGAT input data with connected entities: $h(head)$ and $t(tail)$, and their connection $r(relation)$. Both cultural heritage and attribute information can be h or t , with r describing the connection. All items are represented as nodes and embedded as vectors in R^d , while all relations are edges and embedded as vectors in R^k . The embedding dimensions d, k are set to 256.

2.3 Recommender System

The goal of this algorithm is to analyze relationships between cultural heritage items and attribute, recommending similar items to visitors based on cosine similarity. To this end, we utilize KGAT [2, 3]. The system learns embedding vectors $e_h, e_t, e_r \in R^d$ for each node and relation in the knowledge graph G . To train the model, we define a cost function g by

$$g(h, r, t) = \|W_r e_h + e_r - W_r e_t\|_2^2. \quad (1)$$

When nodes h and t are connected by relation r , these embeddings are trained to minimize $g(h, r, t)$ and when there are no such connection, the embeddings are trained to maximize $g(h, r, t')$. The final loss function is defined as follows:

$$L_{kg} = \sum_{(h,r,t) \in G, (h,r,t') \notin G} -\log \sigma(g(h, r, t) - g(h, r, t')).$$

TABLE I. CULTURAL HERITAGE ONTOLOGY DATASET

Index	Minor cate gory	Place	Cate gory 1	Cate gory 2	Cate gory 3
1	Painting	NMK	Tree	Mountain	Cow
2	Painting	NMK	Mountain	-	-
3	Painting	NMK	Mountain	Tree	-
4	Painting	NMK	Swallow	-	-

The linear transformation $W_r \in R^{d \times d}$ is a learnable parameter, and the sigmoid function $\sigma(x)$ is defined as $1/(1 + e^{-x})$.

2.4 Cold Start Problem

We propose a zero shot embedding method for a new cultural heritage item with attribute information. We utilize the pre-learned embedding vectors for attributes and relationships. This can be done by solving a least squares problem. For a new item $\hat{h}$ with $\{(\hat{h}, r_i, t_i)\}_{i=1}^N$ triples in the graph, we want to find an embedding that solves

$$\min_{e_{\hat{h}}} L = \min_{e_{\hat{h}}} \sum_{i=1}^N \|W_r e_{\hat{h}} + e_r - W_r e_{t_i}\|_2^2.$$

Simply computing the first order condition yields:

$$\frac{\partial L}{\partial e_{\hat{h}}} = \sum_{i=1}^N 2W_{r_i}^T W_{r_i} e_{\hat{h}} - \sum_{i=1}^N 2W_{r_i}^T (W_{r_i} e_{t_i} - e_{r_i}) = 0,$$

$$e_{\hat{h}} = \left(\sum_{i=1}^N W_{r_i}^T W_{r_i} \right)^{-1} \left(\sum_{i=1}^N W_{r_i}^T (W_{r_i} e_{t_i} - e_{r_i}) \right).$$

By doing so, the embedding vector for the new cultural heritage can be obtained without additional training, and the recommender system can be applied.

III. EXPERIMENTS AND RESULTS

Our recommender system recommends the most relevant yet unvisited cultural heritages using the ontology data shown in Table I and addresses the cold start problem for new cultural heritage items.

In the recommendation results based on the similarity of embedding vectors, items from the same major category are often recommended. When items from different major categories are recommended (shown in blue in Table II, it is usually due to similar attributes as we can see in Table III.

TABLE II. SOME EXAMPLES OF RECOMMENDATION RESULT

Start	Recommendation			
	1	2	3	4
Painting 2499	Painting 2411	Painting 2124	Wooden furniture 454	Wooden furniture 461
Painting 2500	Painting 2471	Painting 1672	Painting 336	Painting 2307
Wooden furniture 110	Wooden furniture 103	Wooden furniture 349	Wooden furniture 133	Accessory 806
Wooden furniture 111	Painting 1753	Painting 735	Painting 2446	Painting 1653

TABLE III. ATTRIBUTE INFORMATION OF WOODEN FURNITURE 111 AND PAINTING 1753

Wooden furniture 111	Wardrobe	Dear	Flower	Pine	Crane
Painting 1753	Painting	Dear	Flower	Pine	person

TABLE IV. NEW CULTURAL HERITAGE ATTRIBUTE INFORMATION

name	Minor category	Categor y1	Categor y2	Categor y3	Categor y4
Painting 1712	Painting	Butterfly	Mouse	Egg plant	Frog
New	Painting	Butterfly	Mouse	Egg plant	Frog

TABLE V. NEW CULTURAL HERITAGE ATTRIBUTE INFORMATION

Name	Painting 1712	New
Mean of (1)	3.5166	0.897

To verify that the cold start problem for new cultural heritage items is well addressed, we create a synthetic cultural heritage item with the same attribute information as an existing item (Table IV).

By obtaining the embedding vectors using the proposed method, we can see in Table V that the entities are well represented in the embedding without additional training. The embedding vector of a new cultural heritage item can also be used to recommend other cultural heritage items.

IV. CONCLUSION

In this study, we designed a knowledge graph using cultural heritage ontology data and developed a cultural heritage recommender system capable of making various recommendations based on similarity of embedding vectors of cultural heritage and attribute information. Additionally, we proposed a method to solve the cold start problem for new item by solving the least square problem with embedding vectors. Through the algorithm proposed in this study, users can have a better experience. We believe by combining user interaction data with cultural heritage data can help developing a more precise recommendation system.

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